

Sydney Girls High School



Trial Higher School Certificate

2001

Mathematics

Extension 1

Time Allowed – 2 hours
(Plus 5 minutes reading time)

Directions to Candidates

Name _____

- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Board-approved calculators may be used
- Each question attempted should be started on a new sheet. Write on one side of the paper only

This is a trial paper ONLY. It does not necessarily reflect the format or the contents of the 2001 HSC Examination Paper in this subject.

Question 1

(a) Solve $\frac{4}{x-1} < 2$

(b) Differentiate $y = \tan^{-1} 4x$

(c) Find the coordinates of the point which divides the interval PQ where $P = (2, 5)$ and $Q = (6, 2)$ externally in the ratio 1:3

(d) Evaluate $\int_{-1}^0 2x \sqrt{1+x} \, dx$ using the substitution $u = 1+x$

(e) Find $\int_1^2 \frac{4}{\sqrt{4-x}^2} \, dx$

Question 2

- (a) The polynomial $x^3 + mx^2 + nx - 18$ has $(x + 2)$ as one of its factors. Given that the remainder is -24 when the polynomial is divided by $(x - 1)$, find constants m and n .

(3)

- (b) A circular disc of radius r cm is heated. The area increases due to expansion at a constant rate of 3.2 cm^2 per minute. Find the rate of increase of the radius when $r = 20$ cm.

(3)

- (c) Solve the equation $\sin 2\theta = 2 \sin^2 \theta$

$$\text{for } 0 \leq \theta \leq 2\pi$$

(3)

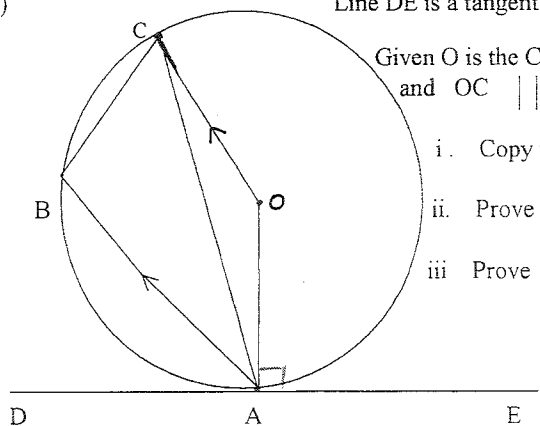
- (d) For the function $y = 3 \sin^{-1} \frac{x}{2}$

- State the domain and range
- Sketch the graph of this function

(3)

Marks

Question 3

(a)  Line DE is a tangent to the Circle at Po
Given O is the Centre of the Circle
and $OC \parallel AB$

- Copy this diagram
- Prove $\angle CAD = \angle E$
- Prove $\angle CBA = 90^\circ$

- (b) Points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$

- Find the equation of chord PQ
- If PQ subtends a right angle at the origin, show that $pq = -4$
- Find the equation of the locus of the midpoint of PQ

- (c) Taking a first approximation of $x = 0.6$ solve the equation $\tan x = x$ using 1 application of Newton's approximation.

Question 4

Marks

- (a) For $y = 10^x$, find $\frac{dy}{dx}$ when $x = 1$ (2)
- (b) Prove that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ (2)
- (c) Two roots of the polynomial $x^3 + ax^2 + 15x - 7 = 0$ are equal and rational. Find a (3)
- (d) For a falling object, the rate of change of its velocity is $\frac{dv}{dt} = -k(v - A)$ where k and A are constants. (5)
- Show that $v = A + Ce^{-kt}$ is a solution of the above equation, where $C = \text{constant}$.
 - If $A = 500$ then initial velocity is 0 and velocity when $t = 5$ seconds is 21 m/s. Find C and k
 - Find the velocity when $t = 20$ seconds
 - Find the maximum velocity as t approaches infinity.

Question 5

- (a) Find the term of the expansion $\left(\frac{2}{x^3} - \frac{x}{3}\right)^8$ which is independent of x
- (b) A particle is moving in S.H.M. with acceleration $\frac{d^2x}{dt^2} = -4x \text{ m/s}^2$
- The particle starts at the origin with a velocity of 3 m/s.
- Find
- the period of the motion
 - the amplitude
 - the speed as an exact value when the particle is 1m from the origin
- (c) Prove by mathematical induction that the expression $(13 \times 6^n + 2)$ is divisible by 5 for all positive integers $n \geq 1$
- (d) Solve $\sqrt{3} \sin \theta - \cos \theta = 1$ for $0 \leq \theta \leq 2\pi$

Question 6

Marks

- (a) Find the acute angle between the lines $x + y = 0$ and $x - \sqrt{3}y = 0$

(3)

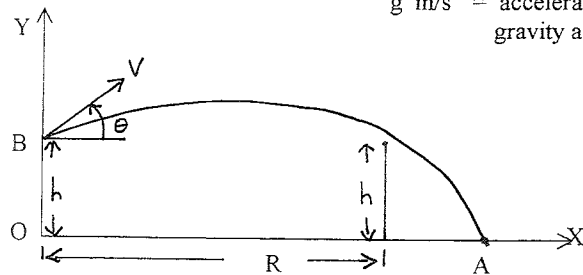
- (b) Show that $\frac{2 \sin^3 x + 2 \cos^3 x}{\sin x + \cos x} = 2 - \sin 2x$

(3)

if $\sin x + \cos x \neq 0$

OC = R metres

$g \text{ m/s}^2$ = acceleration due to gravity acting downwards



A ball is hit from point B which is h metres above the ground level (OX) at an angle of θ from the horizontal level with initial velocity $V \text{ m/s}$. DC represents a fence also of height h metres.

- i. Show that the position of the ball at time t secs is given by

$$x = Vt \cos \theta$$

$$y = Vt \sin \theta - \frac{1}{2}gt^2 + h \quad (2)$$

- ii. Hence show that the equation of flight of the ball is given by

$$y = h + 2 \tan \theta - \frac{x^2 g}{2V^2 \cos^2 \theta} \quad (2)$$

- iii. If the ball clears the fence DC, show that $V^2 \geq \frac{gR}{2 \sin \theta \cos \theta}$

(2)

Question 7

- (a) Use the identity $(1+x)^n = (1+x)(1+x)^{n-1}$ to prove that ${}^n C_r = {}^{n-1} C_{r-1} + {}^{n-1} C_r$

- (b) A car rental company rents 200 cars per day when it sets its hiring rate at \$30 per car for each day.

For every \$1 increase in the hiring rate, 5 fewer cars are rented per day.

- What rate will produce the maximum income per day?
- Find the maximum possible income per day.

- (c) On a building construction site, an object falls from a crane in a vertical straight line. The object passes a 2 metre high window in a time interval of one tenth of 1 second.

Find the height above the top of the window from which the object was dropped

(Take $g = 9.8 \text{ ms}^{-2}$)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

$$\frac{4}{x-1} < 2$$

1st critical value is $x = 1$
(c.v.)

$$\text{Let } \frac{4}{x-1} = 2$$

$$4 = 2x - 2$$

$$6 = 2x$$

$$x = 3 \text{ is 2nd c.v.}$$

$$\leftarrow \leftarrow \rightarrow \rightarrow$$

$$\text{Test } x = 0 \quad \frac{4}{-1} < 2 \text{ True}$$

$$x = 2 \quad \frac{4}{2-1} < 2 \text{ False}$$

$$x = 4 \quad \frac{4}{3} < 2 \text{ True}$$

$$\text{Ans: } x < 1, x > 3$$

$$P = (2, 5) = x_1, y_1$$

$$Q = (6, 3) = x_2, y_2$$

$$k_1 : k_2 = 1 : -3$$

$$x = \frac{k_1 x_2 + k_2 x_1}{k_1 + k_2} = \frac{1 \times 6 - 3 \times 2}{1 - 3}$$

$$= 0$$

$$y = \frac{k_1 y_2 + k_2 y_1}{k_1 + k_2} = \frac{1 \times 3 - 3 \times 5}{1 - 3}$$

$$= \frac{-12}{-2} = 6$$

$$\text{Ans} = (0, 6)$$

$$\int_1^2 \frac{4}{\sqrt{4-x^2}} dx$$

$$= 4 \int_1^2 \frac{1}{\sqrt{2^2 - x^2}} dx$$

$$= 4 \left[\sin^{-1} \left(\frac{x}{2} \right) \right]_1^2$$

$$4 \left[\sin^{-1}(1) - \sin^{-1} \left(\frac{1}{2} \right) \right]$$

$$4 \left[\frac{\pi}{2} - \frac{\pi}{6} \right] = \frac{4}{3} \pi$$

$$\frac{6}{y} = \tan^{-1}(4x)$$

$$\text{Let } u = 4x \quad \therefore y = \tan^{-1} u$$

$$\frac{du}{dx} = 4 \quad \frac{dy}{du} = \frac{1}{1+u^2}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= \frac{1}{1+u^2} \cdot 4$$

$$= \frac{4}{1+16x^2}$$

$$\frac{dI}{dx} = \int_{-1}^0 2x \sqrt{1+x} dx$$

$$\text{Let } u = 1+x$$

$$x = -1, u = 0$$

$$x = 0, u = 1$$

$$2x \sqrt{1+x}$$

$$= 2(u-1)u^{1/2}$$

$$= 2(u^{3/2} - u^{1/2})$$

$$\frac{du}{dx} = 1$$

$$\therefore du = dx$$

$$I = 2 \int_0^1 u^{3/2} - u^{1/2} du$$

$$= 2 \left[\frac{2u^{5/2}}{5} - \frac{2u^{3/2}}{3} \right]_0^1$$

$$= 2 \left[\frac{2}{5} - \frac{2}{3} \right]$$

$$= -\frac{8}{15}$$

$$L = m + n$$

Answer 2

$$(a) P(x) = x^3 + mx^2 + nx - 18$$

$$P(-2) = -8 + 4m - 2n - 18$$

$$= 4m - 2n - 26 = 0 \quad (*)$$

$$P(1) = 1 + m + n - 18 = -24$$

$$m + n + 7 = 0 \quad (**)$$

Solve simultaneously

$$2m + 2n + 14 = 0$$

$$6m - 12 = 0$$

$$\therefore m = 2$$

$$n = -9$$

$$(c) \sin 2\theta = 2 \sin^2 \theta, (0 \leq \theta \leq 2\pi)$$

$$2 \sin \theta \cos \theta = 2 \sin^2 \theta$$

$$2 \sin \theta (\sin \theta - \cos \theta) = 0$$

$$\therefore \sin \theta = 0 \text{ or } \sin \theta = \cos \theta$$

$$\tan \theta = 1$$

$$\theta = 0^\circ, \pi, 2\pi, \frac{\pi}{4}, \frac{5\pi}{4}$$

$$(b) A = \pi r^2$$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \frac{dr}{dt}$$

$$3.2 = 2\pi r \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = \frac{3.2}{2\pi r}$$

$$= 0$$

Rate of incr.

is 0.

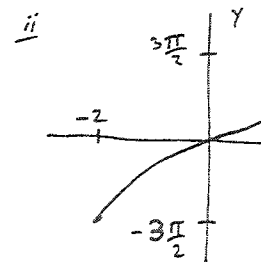
$$(d) y = 3 \sin x$$

$$-1 \leq \frac{y}{3}$$

$\therefore$ domain

$$-\frac{\pi}{2} \leq \sin x \leq \frac{\pi}{2}$$

$$\text{Range } -\frac{3\pi}{2} \leq y$$



Question 4

$$\frac{d}{dt} \ddot{v} = A + C e^{-kt}$$

$$0 = 500 + C e^0$$

$$\therefore C = -500$$

$$21 = 500 - 500 e^{-5k}$$

$$500 e^{-5k} = 479$$

$$e^{-5k} = \frac{479}{500}$$

$$-5k \log e = \log e \left(\frac{479}{500} \right)$$

$$\therefore k = 0.0085815$$

$$\text{iii} \quad v = 500 - 500 e^{-0.0085815 \times 20} = 78.9 \text{ m/s}$$

$$\text{iv} \quad v = 500 - \frac{500}{e^{0.0085815 \times t}}$$

$$\text{as } t \rightarrow \infty, v \rightarrow 500 \text{ m/s}$$

$$\therefore \text{max Velocity} = 500 \text{ m/s}$$

Question 5

$$(a) \left(\frac{2}{x^3} - \frac{x}{3} \right)^8$$

$$T_{k+1} = {}^nC_k a^{n-k} b^k$$

$$= {}^8C_k \left(\frac{2}{x^3} \right)^{8-k} \left(-\frac{x}{3} \right)^k$$

$$= {}^8C_k \frac{2^{8-k}}{x^{24-3k}} \cdot (-1)^k \frac{x^k}{3^k}$$

$$= {}^8C_k \frac{2^{8-k}}{3^k} (-1)^k x^{4k-24}$$

For term independent of x

$$4k - 24 = 0$$

$$k = 6$$

$$\therefore T_7 = (-1)^6 {}^8C_6 \frac{2^2}{3^6}$$

$$= \frac{112}{729} = \text{Term indep. of } x.$$

$$(c) \text{ Put } n = 1$$

$$13 \times 6^n + 2 = 13 \times 6^1 + 2 = 80$$

This is divisible by 5

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$13 \times 6^k + 2 = 5m, \text{ for integer } m$$

Prove true for $n = k+1$

$$13 \times 6^{k+1} + 2 = 6(13 \times 6^k + 2) - 10$$

$$= 6(5m) - 5 \times 2$$

$$= 5(6m - 2)$$

This is divisible by 5.

$\therefore$ True for $n = k+1$

If the result is true for $n = k$

Then it is true for $n = k+1$

Since it is true for $n = 1$, then it is true for $n = 2, n = 3$ etc.

$$(b) \ddot{x} = -4x = -\pi^2 x$$

$$\therefore n = 2$$

$$\text{Period: } T = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi$$

$$b \quad v^2 = n^2(a^2 - x^2)$$

$$3^2 = 2^2(a^2 - 0^2)$$

$$\therefore a = 3/2 = 1.5 = \text{amplitude.}$$

$$c \quad v^2 = n^2(a^2 - x^2)$$

$$= 2^2 \left(\frac{3^2}{2^2} - 1^2 \right)$$

$$= 4 \left(\frac{9}{4} - 1 \right)$$

$$= 9 - 4$$

$$= 5$$

$$\therefore v = \pm \sqrt{5} \text{ m/s}$$

$$d \quad \sqrt{3} \sin \theta - \cos \theta = 1 \quad \text{or } 0$$

$$\text{Using } t = \tan \frac{\theta}{2}$$

$$\sin \theta = \frac{2t}{1+t^2} \quad \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\frac{2\sqrt{3}t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$$

$$2\sqrt{3}t - 1 + t^2 = 1+t^2$$

$$t = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\tan \frac{\theta}{2} = \frac{1}{\sqrt{3}}$$

$$\frac{\theta}{2} = \frac{\pi}{6}$$

$$\therefore \theta = \frac{\pi}{3}$$

Also test $\theta = \pi$ since t method won't prove this

$$\sqrt{3} \sin \pi - \cos \pi = -(-1)$$

$\therefore \theta = \pi$ is a solution

$$\text{Ans: } \theta = \frac{\pi}{3} \text{ and } \pi$$

$$\begin{aligned}
 x - \sqrt{3}y &= 0 \\
 y &= -\frac{1}{\sqrt{3}}x \\
 \therefore m_1 &= -1 \\
 m_2 &= \frac{1}{\sqrt{3}}
 \end{aligned}$$

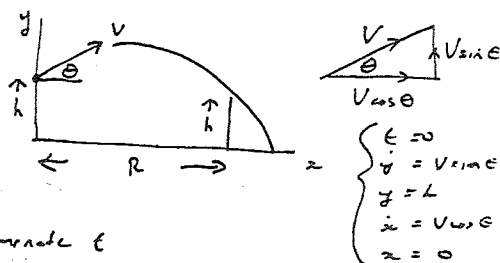
$$\begin{aligned}
 \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\
 &= \left| \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}} \right| \times \frac{\sqrt{3}}{\sqrt{3}} \\
 &= \frac{1 + \sqrt{3}}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} \\
 &= \frac{1 + 3 + 2\sqrt{3}}{2} \\
 &= 2 + \sqrt{3}
 \end{aligned}$$

$$\therefore \theta = 75^\circ$$

$\therefore$ i Consider vertical motion
up being +

$$\begin{aligned}
 \ddot{y} &= -g \\
 \dot{y} &= -gt + c \\
 V \sin \theta &= c \\
 \dot{y} &= -gt + V \sin \theta \\
 y &= -\frac{gt^2}{2} + Vt \sin \theta + k \\
 k &= h \\
 \therefore y &= Vt \sin \theta - \frac{1}{2}gt^2 + h \\
 \text{Consider horizontal motion} \\
 \ddot{x} &= 0 \\
 \dot{x} &= c \\
 V \cos \theta &= c \\
 \dot{x} &= V \cos \theta \\
 x &= Vt \cos \theta + c \\
 c &= 0 \\
 \therefore x &= Vt \cos \theta
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \frac{2(\sin^3 x + \cos^3 x)}{\sin x + \cos x} \\
 &= \frac{2(\sin x + \cos x)(\sin^2 x - \sin x \cos x + \cos^2 x)}{(\sin x + \cos x)} \\
 &= \frac{2(1 - \sin x \cos x)}{1} \\
 &= 2 - 2 \sin x \cos x \\
 &= 2 - \sin 2x \quad (\text{if } \sin x + \cos x \neq 0)
 \end{aligned}$$



ii Eliminate t
 $t = \frac{x}{V \cos \theta}$

$$\begin{aligned}
 y &= h + \frac{V \sin \theta x}{V \cos \theta} - \frac{g x^2}{2 V^2 \cos^2 \theta} \\
 y &= h + x \tan \theta - \frac{x^2 g}{2 V^2 \cos^2 \theta}
 \end{aligned}$$

iii For ball to clear the fence
 $x = R \quad y > h$

$$h + R \tan \theta - \frac{R^2 g}{2 V^2 \cos^2 \theta} > h$$

$$R \tan \theta > \frac{R^2 g}{2 V^2 \cos^2 \theta}$$

$$2 V^2 \cos^2 \theta > \frac{R g}{\tan \theta}$$

$$V^2 > \frac{g R}{2 \frac{\sin \theta}{\cos \theta} \cos^2 \theta}$$

$$\therefore V^2 > \frac{g R}{2 \sin \theta \cos \theta}$$

Question 7

$$(1+x)^n = 1 + {}^nC_1 x + \dots + {}^nC_r x^r + \dots + x^n \quad (1)$$

$$(1+x)(1+x)^{n-1} = (1+x)(1 + {}^{n-1}C_1 x + \dots + {}^{n-1}C_{r-1} x^{r-1} + {}^{n-1}C_r x^r + \dots + x^{n-1})$$

$$= (1 + \dots + {}^{n-1}C_r x^r + \dots + x^{n-1}) + (x + \dots + {}^{n-1}C_{r-1} x^r + \dots + x^n) \quad (2)$$

Equating Coefficient of x^r in Line (1) with Coeff of x^r in Line (2)

$$\therefore {}^nC_r = {}^{n-1}C_r + {}^{n-1}C_{r-1}$$

b i Income I = Number of cars rental $\times$ Rate per car per day

Let $\$x$ = additional amount over $\$30$

$$I = (200 - 5x) \cdot (30 + x)$$

$$= 6000 + 200x - 150x - 5x^2$$

$$I = 6000 + 50x - 5x^2$$

$$\frac{dI}{dx} = 50 - 10x$$

$$\frac{d^2I}{dx^2} = -10 < 0 \quad \therefore \text{Max } I$$

Now for maximum I , $\frac{dI}{dx} = 0$

$$50 - 10x = 0$$

$$\therefore x = 5$$

Thus the rate which produces Maximum daily Income

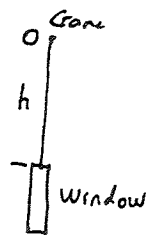
$$= \$30 + \$5 = \$35 \text{ per car per day.}$$

$$\text{ii Maximum Income} = 6000 + (50 \times 5) - 5 \times 5^2$$

$$= \$6125$$

Question 7

c) Consider the downward direction as positive $\downarrow$



Let h = height of crane above top of window

Total Vertical motion $t=0, y=0,$

$$\ddot{y} = +g$$

$$\dot{y} = gt + c$$

$$0 = 0 + c$$

$$\dot{y} = gt$$

$$y = \frac{gt^2}{2} + c$$

$$0 = 0 + c$$

$$\therefore y = \frac{gt^2}{2}$$

Let $t = T$ secs to reach top of window

Velocity at top of window $\dot{y} = gT$

Displacement at top of window $= h = \frac{g}{2} T^2$

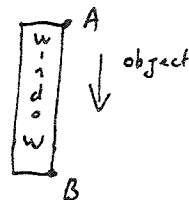
$$\therefore \frac{2h}{g} = T^2$$

Time to reach top of window $= T = \sqrt{\frac{2h}{g}}$

$\therefore$ Vel at top of window $= g \sqrt{\frac{2h}{g}} =$

Now consider Motion from top to bottom of window

Let $t=0, y=0, \dot{y} = \sqrt{2gh}$ at A



$$\ddot{y} = g$$

$$\dot{y} = gt + c$$

$$\sqrt{2gh} = 0 + c$$

$$\dot{y} = gt + \sqrt{2gh}$$

$$y = \frac{gt^2}{2} + \sqrt{2gh} \cdot t + c$$

$$0 = 0 + 0 + c$$

$$y = \frac{gt^2}{2} + t\sqrt{2gh}$$

At B, $y=2, t = \frac{1}{10}$

$$2 = \frac{9.8 \times \frac{1}{100}}{2} + \frac{1}{10}$$

$$2 = 0.049 + \frac{1}{10}$$

$$1.951 \times 10 = \sqrt{19.51}$$

$$19.51 = \sqrt{19.51}$$

$$380.6401 = 19$$

$$\therefore h = 19.42$$

Thus the crane

Q 7c

$\downarrow$
 $2m \left[t = \frac{1}{10} sec \right]$

$$\ddot{x} = 9.8$$

$$\dot{x} = 9.8t + C$$

$$\text{at } t=0, \dot{x}=0=C$$

$$\therefore \dot{x} = 9.8t$$

$$x = 4.9t^2 + C_1$$

$$\text{at } t=0, x=0=C_1$$

$$\therefore x = 4.9t^2$$

$$\text{at } t = t+0.1, x = x+2$$

$$\therefore x+2 = 4.9(t+0.1)^2$$

$$\& \text{ sub } x = 4.9t^2, \quad 4.9t^2 + 2 = 4.9(t^2 + 0.2t + 0.01)$$

$$\therefore 4.9t^2 + 2 = 4.9t^2 + 0.98t + 0.049$$

$$\therefore 2 - 0.049 = 0.98t$$

$$\therefore t = \frac{1.951}{0.98} = \frac{195.1}{98}$$

$$\therefore x = 4.9 \left(\frac{195.1}{98} \right)^2 \doteq 19.4$$

$\therefore$ height above window is 19.4 m.